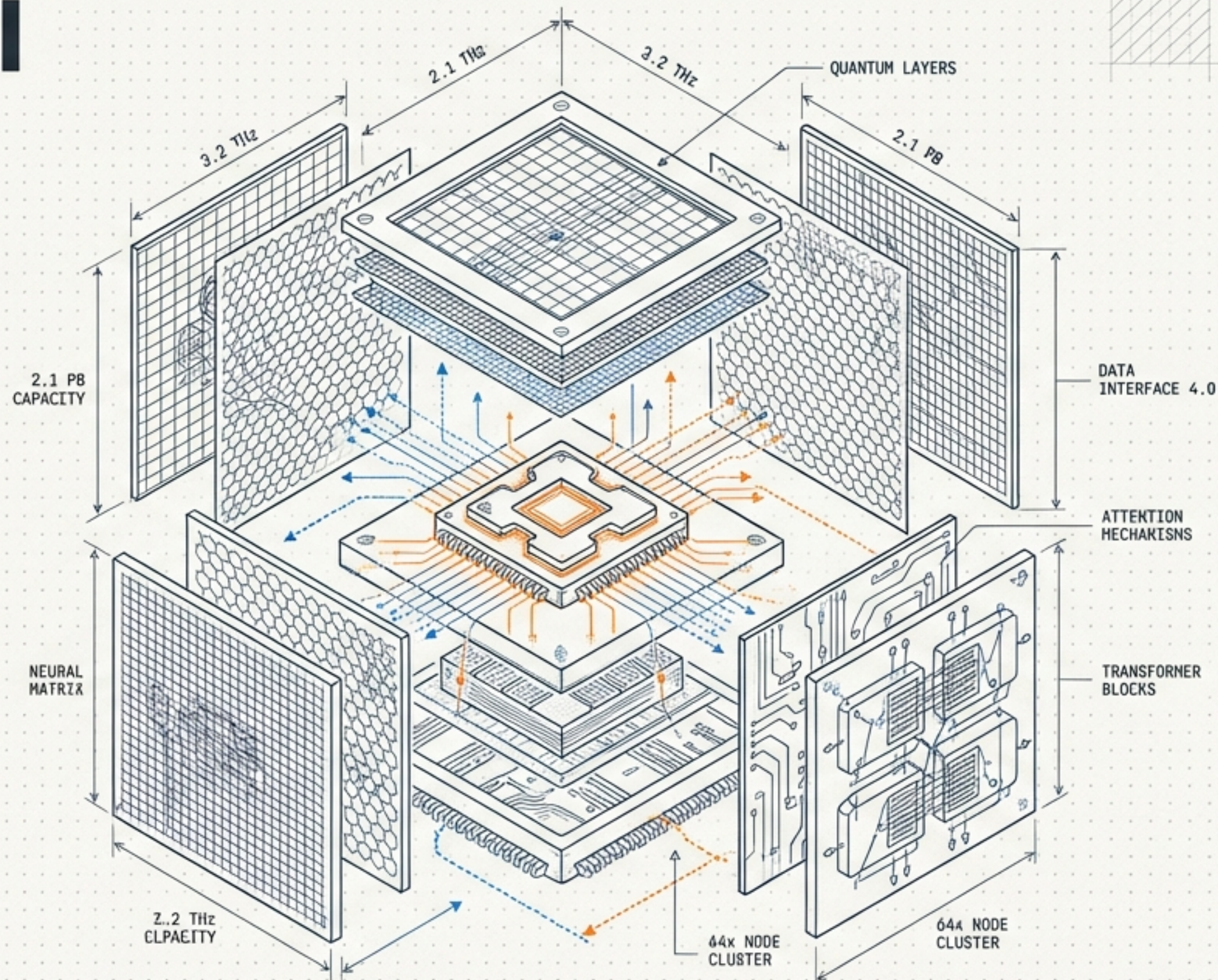


Advanced GenAI Fundamentals

Deconstructing Large Language Models



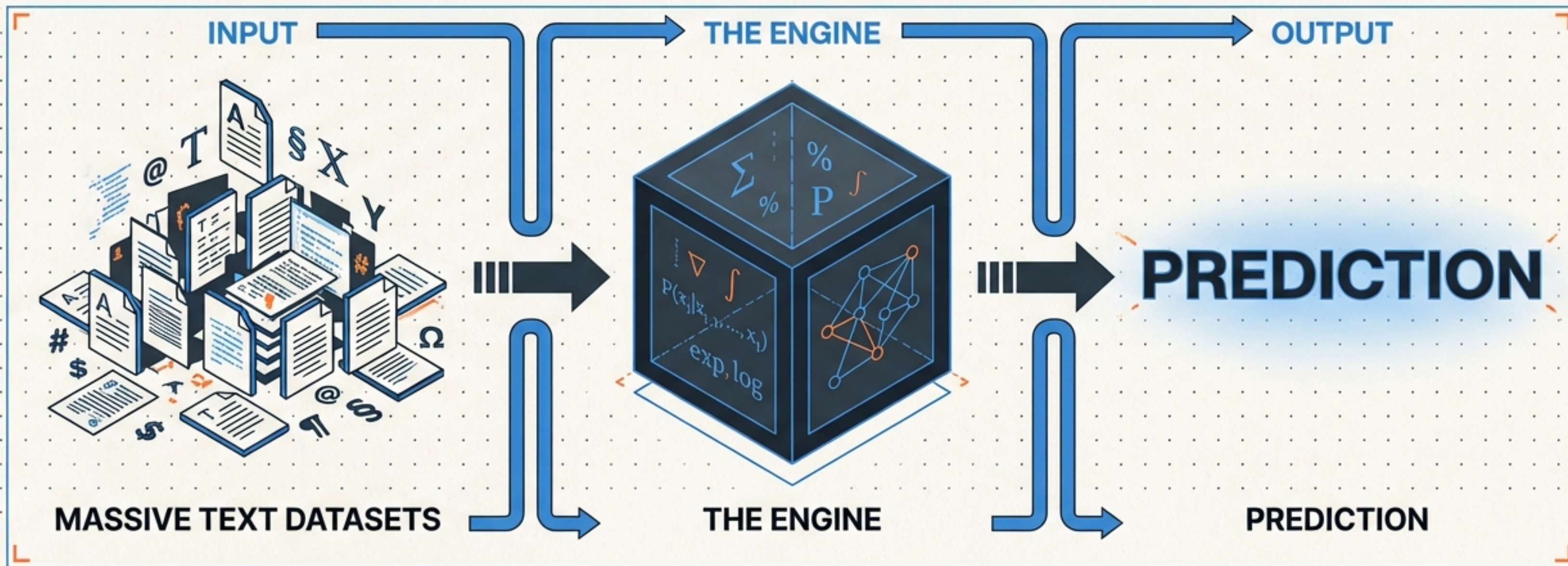
CURRICULUM ID: DAY-12

INTENT: ARCHITECTURE & OPTIMIZATION

ORGANIZATION: SKILLOCEANS

TRAINER: ANSHUL // AIML LEAD

THE PROBABILITY ENGINE



FOUNDATION

Trained on massive corpuses of text data. Powers ChatGPT, GPT-4, and Claude.

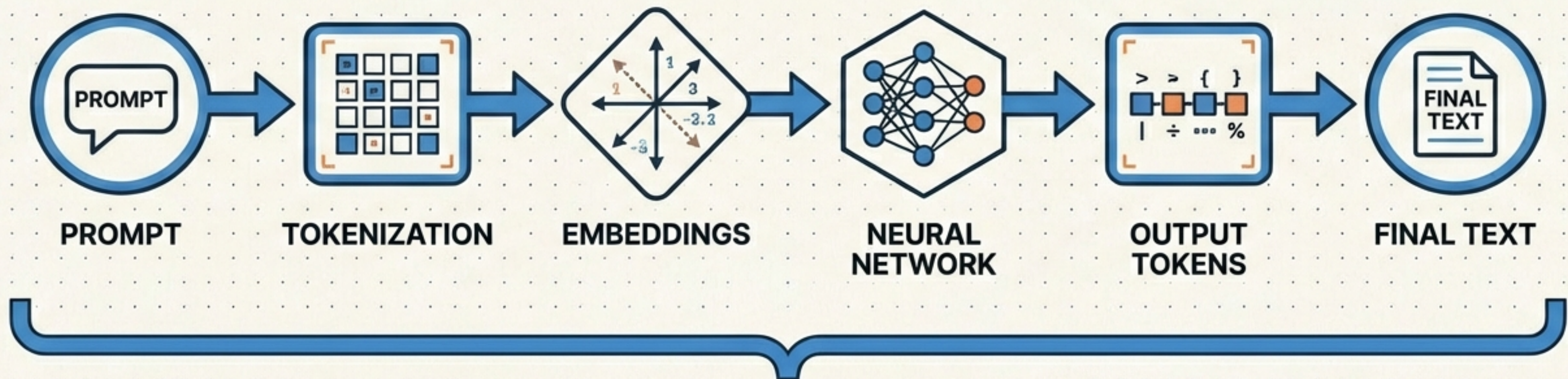
MECHANISM

Operates on statistical probability, not sentience. It predicts the next likely token.

DEFINITION

A system designed to generate coherent sequences based on learned patterns.

THE DATA JOURNEY: PIPELINE ARCHITECTURE

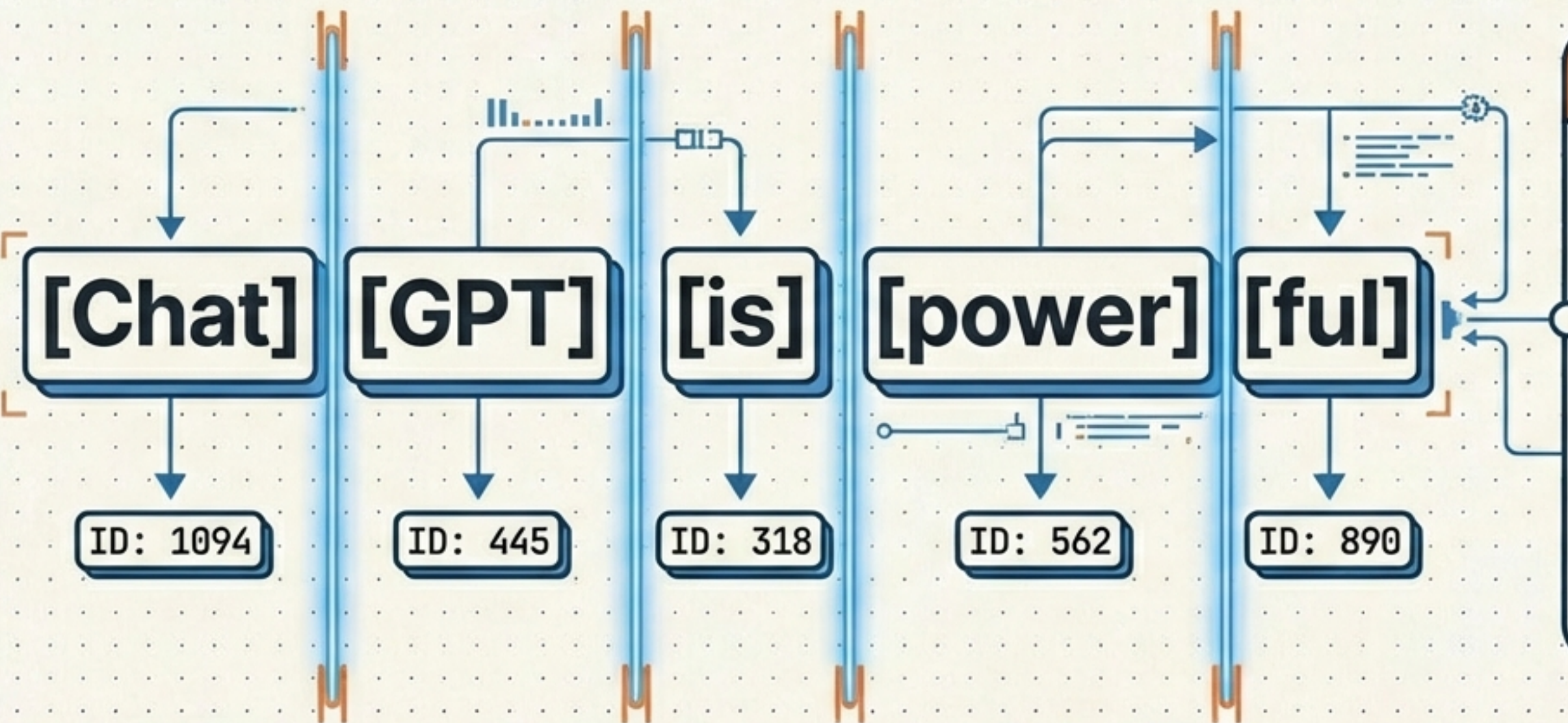


THE LLM PROCESSING FLOW

Every response generated by an LLM follows this exact step-by-step pipeline. The quality of the output is strictly determined by the inputs at Stage 1.

STEP 1: TOKENIZATION

The Atomic Unit of Language



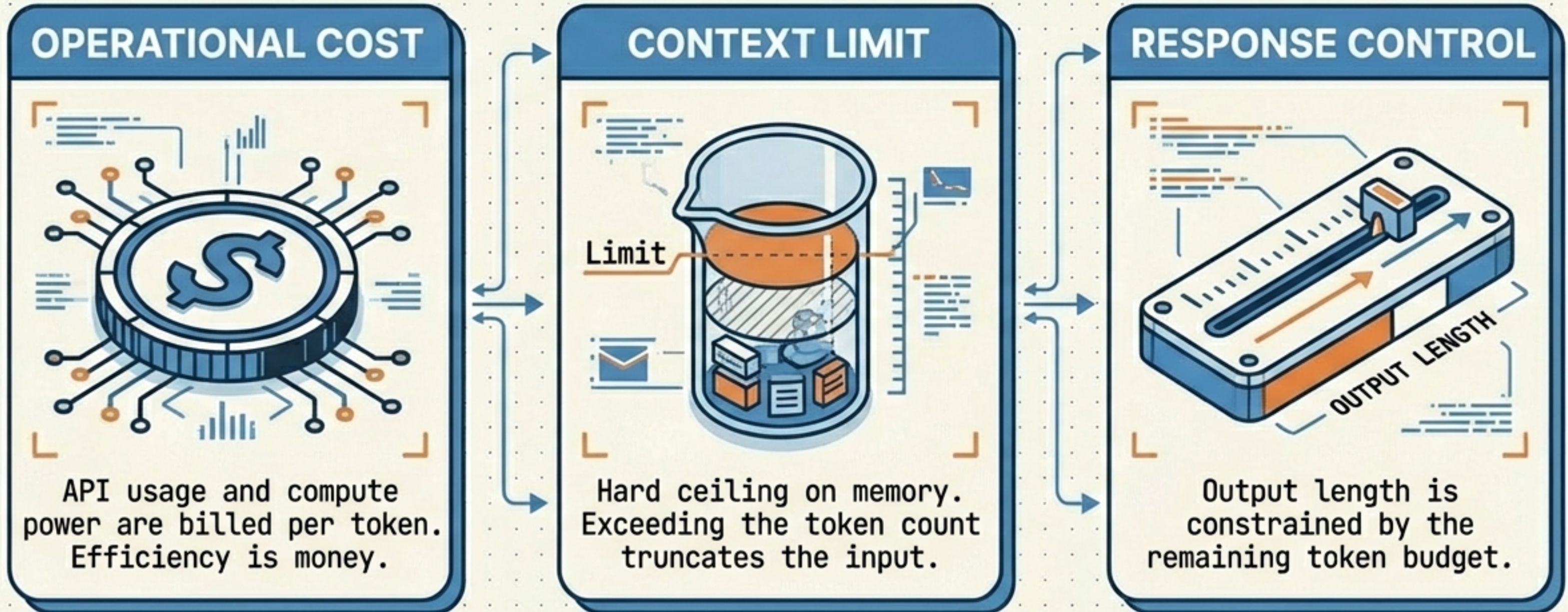
CRITICAL RULE

Words $\neq$ Tokens.

The model does not see text; it sees a sequence of numerical chunks.

Note:
Longer words
= More tokens.

THE ECONOMICS OF TOKENS



Prompt Engineering is essentially Token Budget Management.

STEP 2: EMBEDDINGS

Translating Language into Math



KING

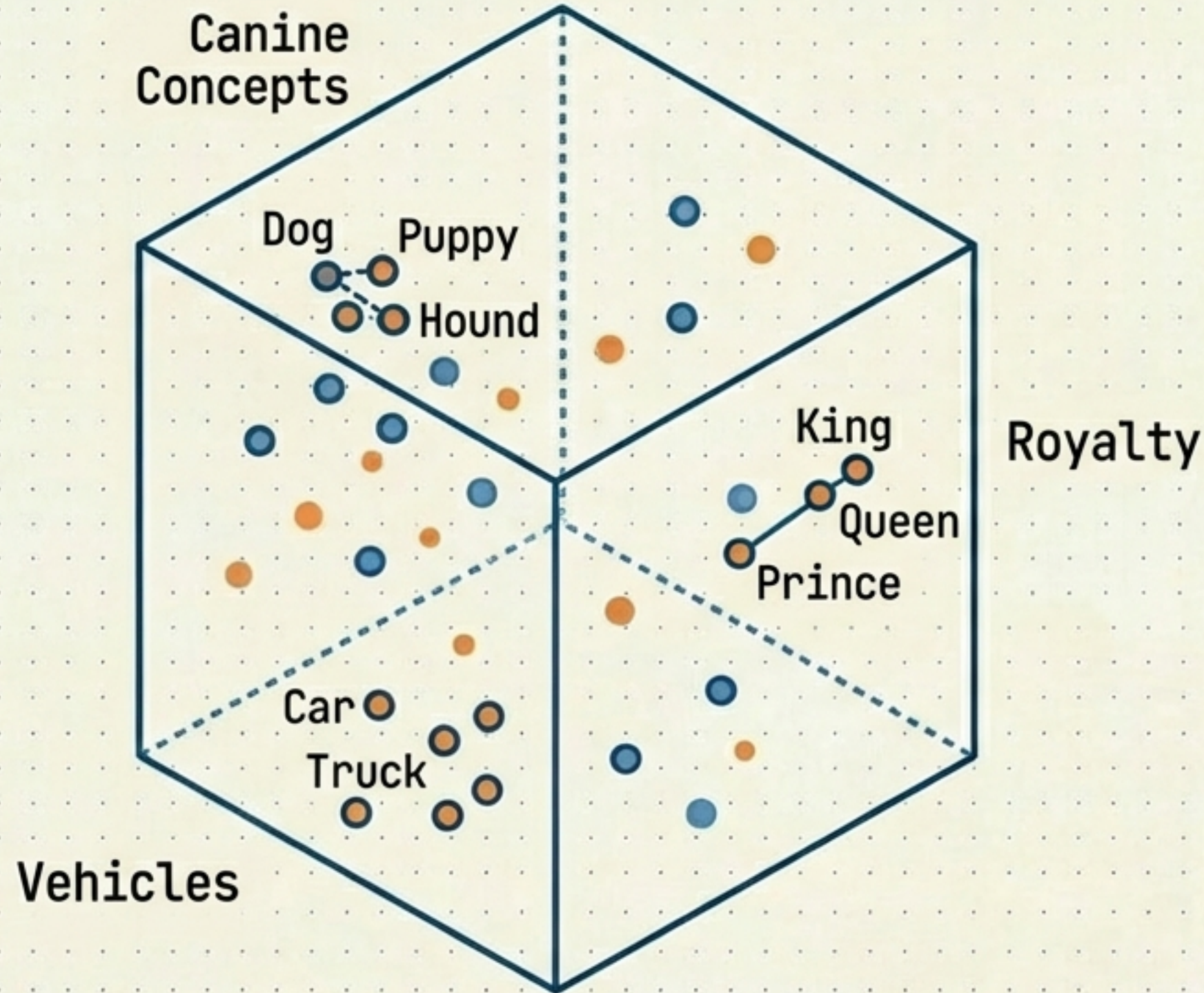
Vectorization



$\begin{bmatrix} 0.82, \\ -0.14, \\ 0.99, \\ 0.45, \\ -0.67, \\ 0.12, \\ 0.88, \\ 0.31, \dots \end{bmatrix}$

- The model operates on numerical coordinates called Vectors.
- These numbers represent semantic meaning, not spelling.
- Rule: Similar Meaning $\rightarrow$ Similar Vector Coordinates.

NAVIGATING THE VECTOR SPACE

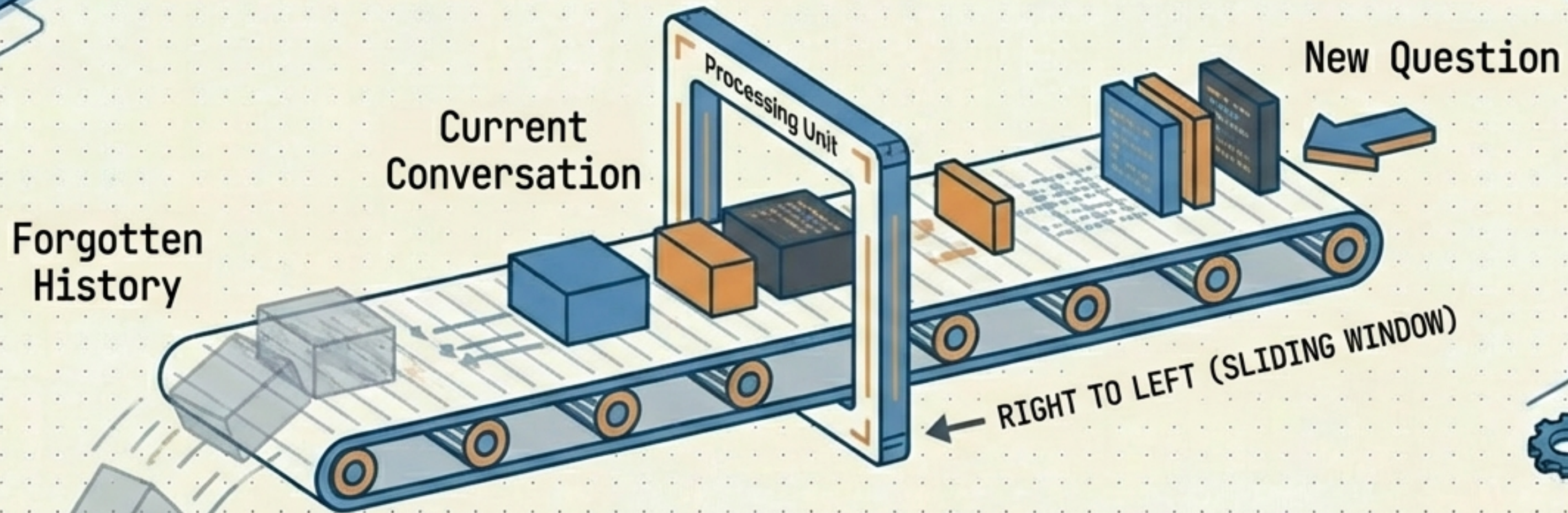


APPLICATIONS:

- Semantic Search (Concept matching)
- Document Similarity
- Recommendation Engines
- Knowledge Base Retrieval

STEP 3: THE CONTEXT WINDOW

Memory Constraints & The Sliding Window

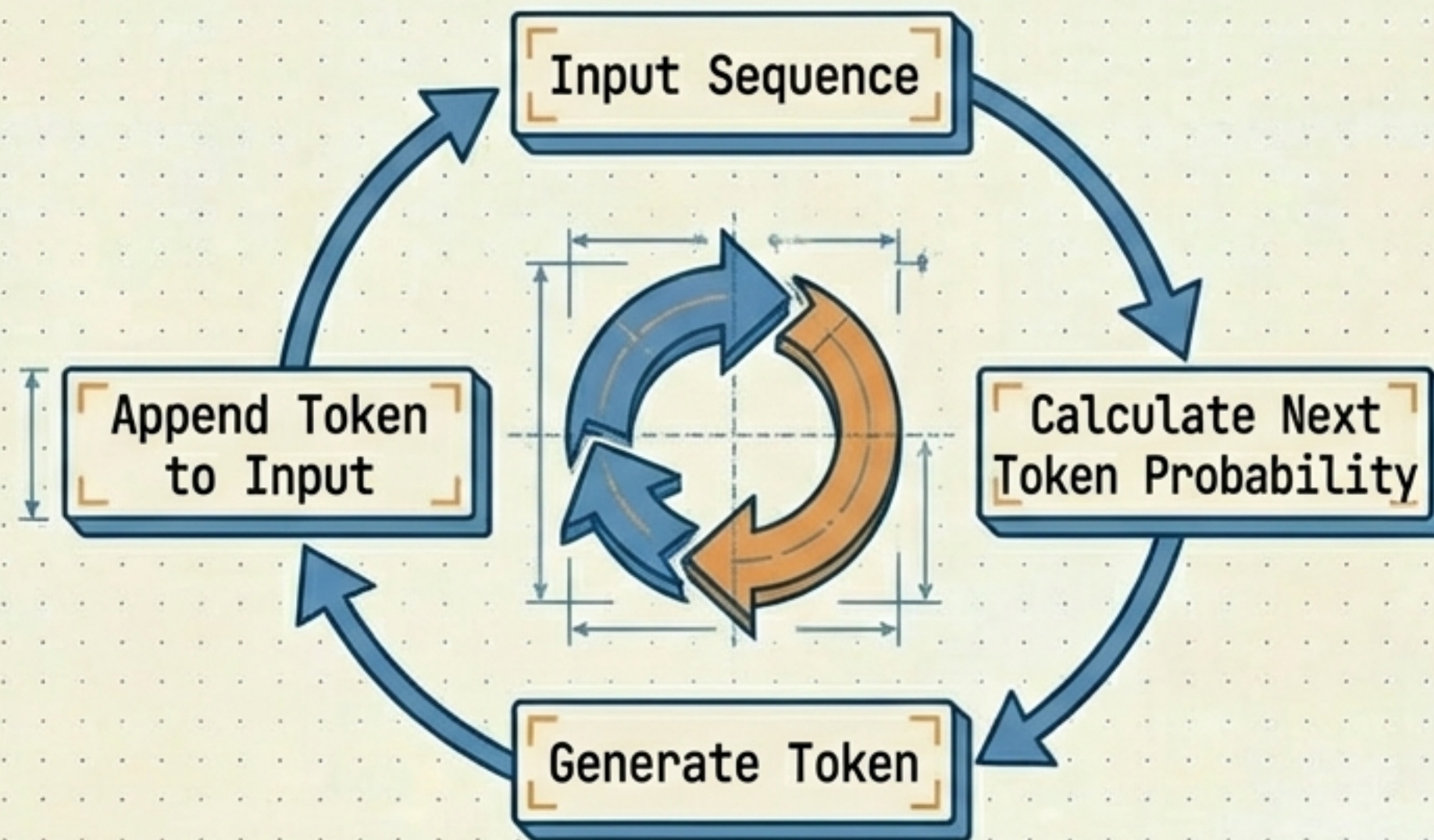


Context Limit = Input Tokens + Output Tokens



FIFO (First-In, First-Out): When the limit is reached, the oldest messages are dropped to make room for new data.

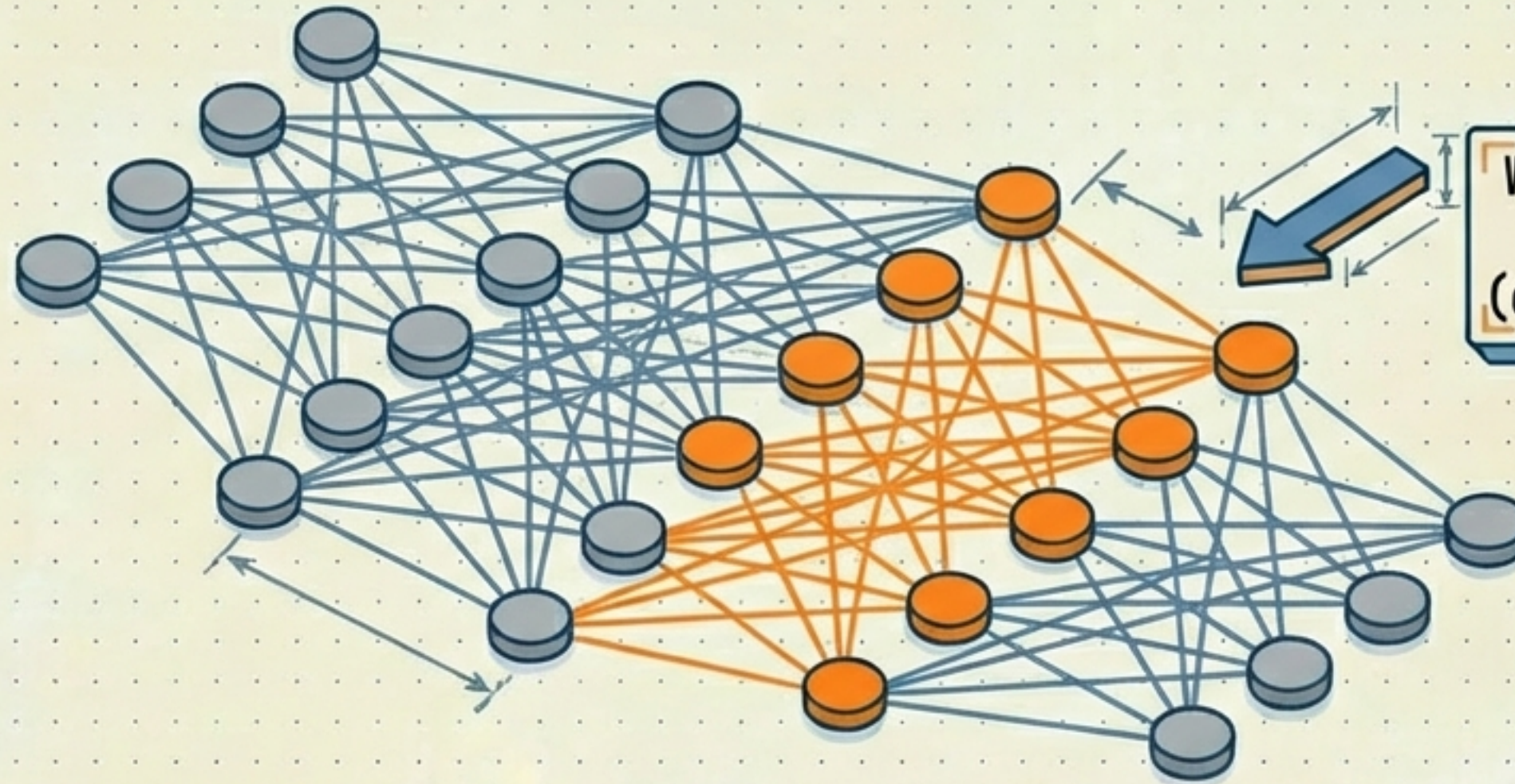
THE PREDICTION LOOP



- 1. The model predicts the single most likely next token.
- 2. That token is added to your prompt.
- 3. The process repeats with the new longer prompt.

This is not a burst of thought. It is an **iterative, stepwise calculation.**

FINE-TUNING: ALTERING THE BRAIN



Weights Adjusted for
Specific Domain
(e.g., Medical Data).

Fine-Tuning is the process of further training a pre-existing model on a custom dataset to permanently alter its weights and biases.

Unlike Prompt Engineering, this changes the model itself, not just the user input.

High Cost / High Accuracy / Specialized Behavior

STRATEGY MATRIX: OPTIMIZATION LEVRS

PROMPT ENGINEERING

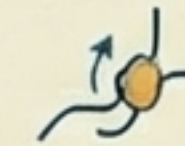
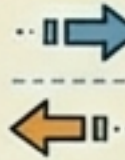


FINE-TUNING

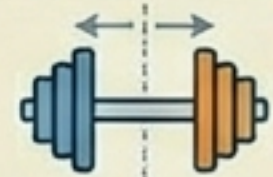


METHOD

No Training Required



Model Training Required



COST

Low (Per Token)



High (Compute Heavy)



SPEED

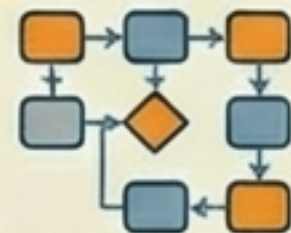
Instant Iteration



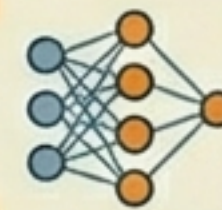
Slow Implementation



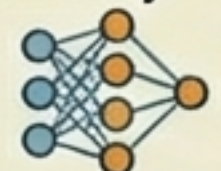
BEST FOR



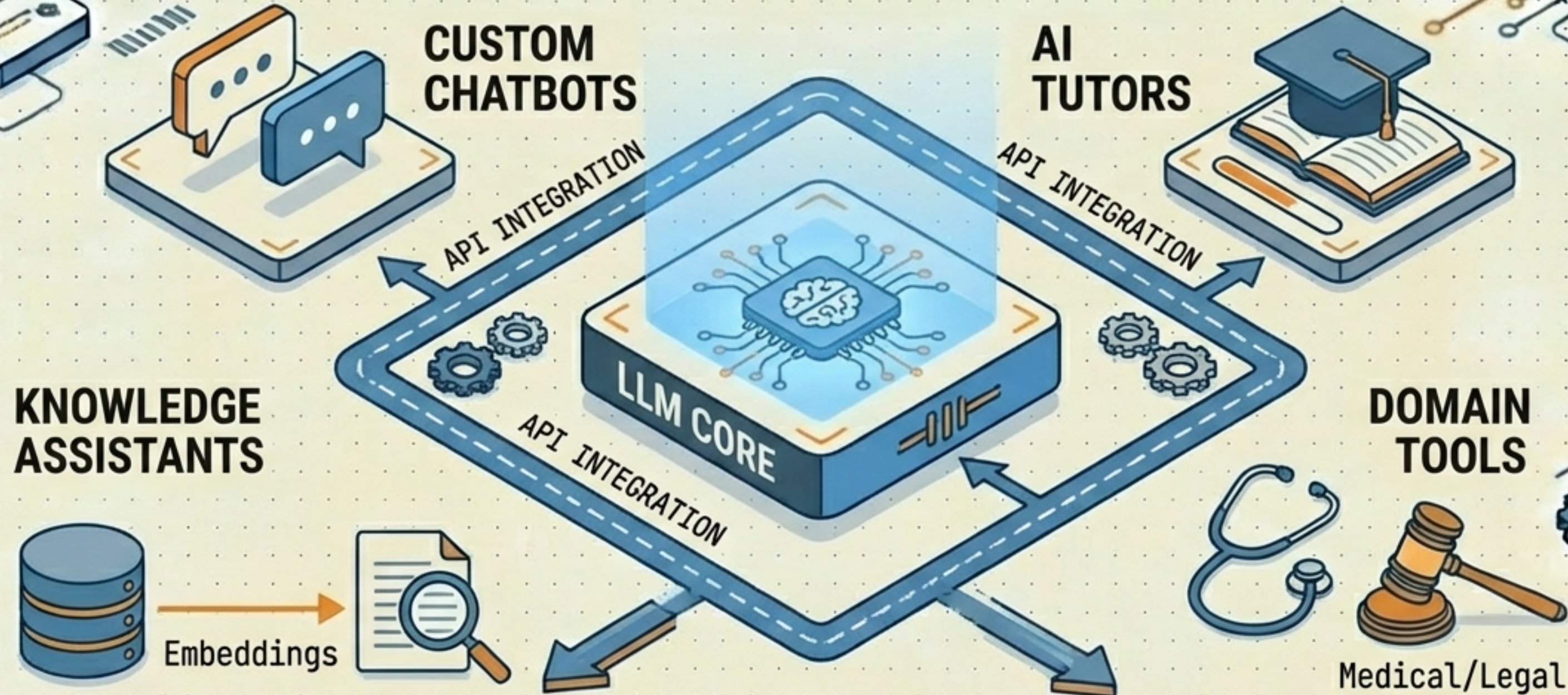
General Tasks,
Prototyping,
Flexible Logic



Deep Domain Expertise,
New Languages,
Specific Styles



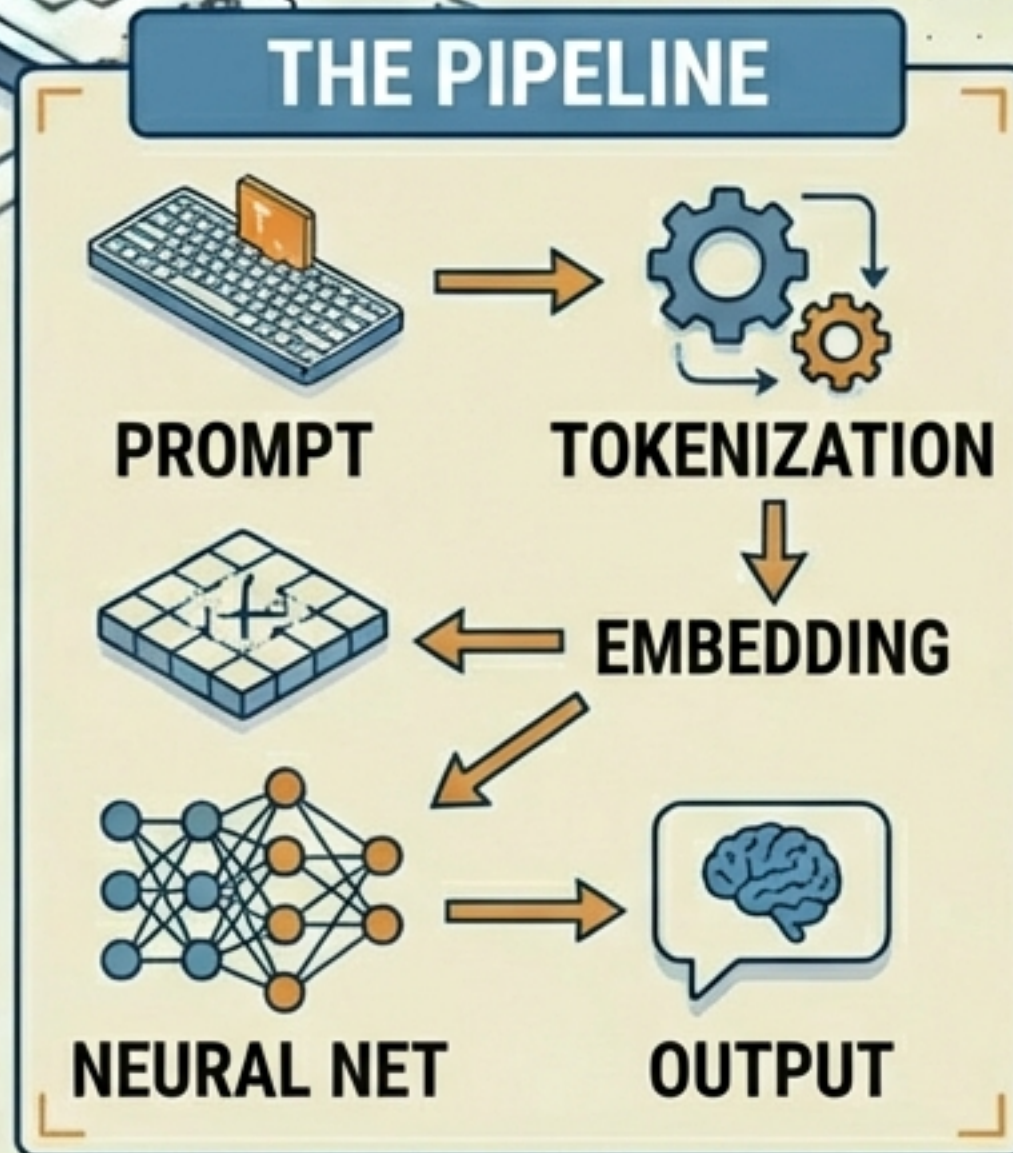
REAL-WORLD ARCHITECTURE



These are not standalone apps. They are specialized pipelines built on top of the Token -> Vector -> Prediction flow.

SYSTEM SUMMARY

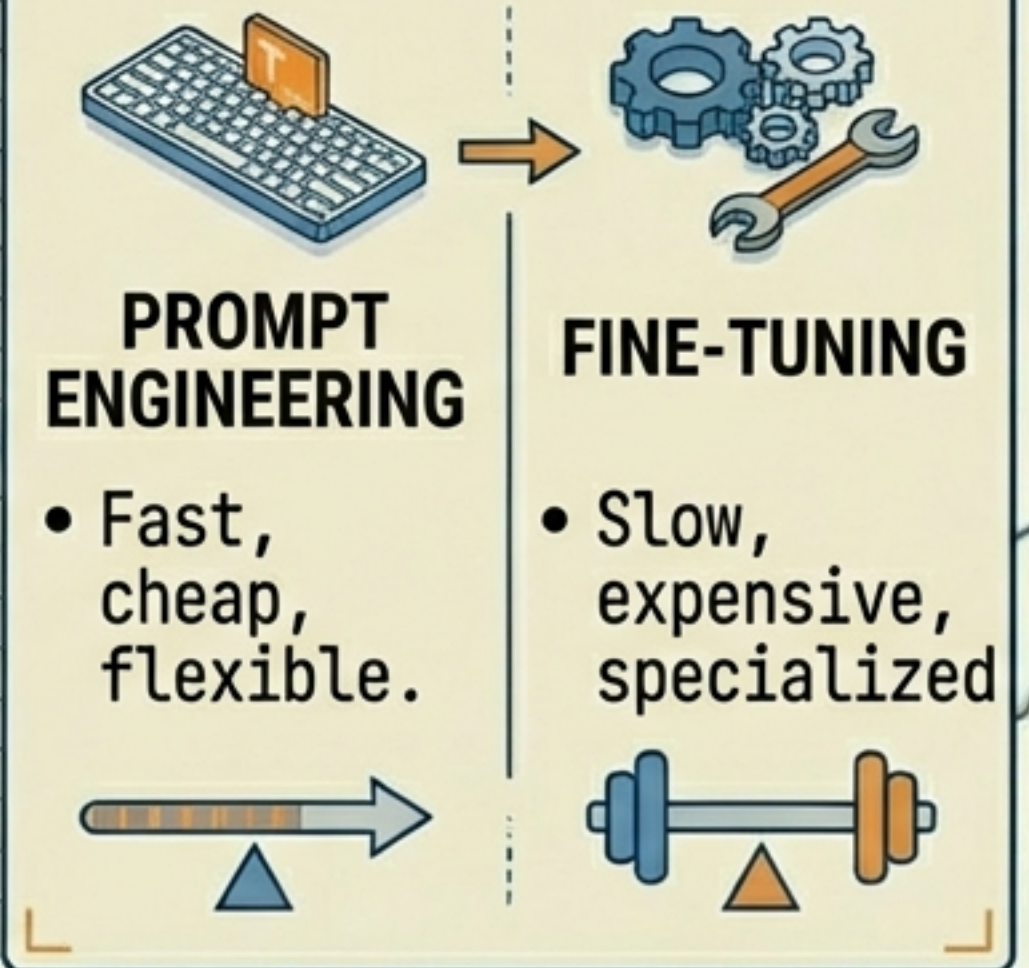
THE PIPELINE



THE MECHANICS

- **Tokens:** The currency of cost and context. (Icon: stack of coins)
- **Vectors:** The mathematical translation of meaning. (Icon: three diverging arrows)
- **Context Window:** The sliding memory limit. (Icon: clock and a slider bar)

THE OPTIMIZATION



Mastering GenAI requires balancing these architectural constraints.

Advanced **GenAI** Fundamentals

Skilloceans Training Series

Trainer: Anshul

Role: AIML Trainer

Contact: [Instructor Email/Link]

[END OF SEQUENCE]